

So, in order to see the shape of the Earth as a whole and to know its unique features, a three dimensional model of the Earth was created with a specific scale.

DO YOU KNOW?

The surface area of the Earth is 510.1 million square kilometres.

The Earth which is spherical, is flat at the poles and bulges at the Equator. The Earth cannot be compared with any other geometrical shape as it has a very unique shape. Hence, its shape is called a **geoid** (earth shaped).

The Earth moves around the Sun. It also rotates from the West to East on its axis at an inclination of 23 ½°. The globe is also inclined at an angle of 23 ½°. The axis is an imaginary line. It is not actually found on the Earth.

DO YOU KNOW?

- The first globe was created by the **Greeks** in the year 150 AD(CE).
- The Indian astronomer **Aryabhatta**-I has mentioned in his book. **'Aryabhatta Sidhantha'**. **'The stars in the sky seem to move towards the West because of the Earth's roation on its axis'.**

Lines on the Globe

There are imaginary lines which are drawn on the globe horizontally and vertically to find a location and calculate distance and time. These imaginary lines are called lines of **latitudes and longitudes.**

DO YOU KNOW?



Ptolemy, a Greco – Roman mathematician, astronomer and geographer, was the first person to draw the lines of latitude and longitude on a map.

In his book, **'Geographia'** a detailed description about the Earth's surface, its size and circumference and many locations based on the lines of latitude and longitude are given.

Latitudes

The imaginary lines which are drawn horizontally on East - West direction on the Earth are called the lines or **parallels** of **latitudes**.

The 0° line of latitude which divides the Earth into two halves is known as the **Equator**. From the Equator, **parallel** lines are drawn towards the North and South poles at equal intervals. The latitudinal extent between **1°** line of latitude on Earth is **111 km**.